

Math F113X: Homework Set 6b — Solutions

Graph Theory: # 19*, 22*, A, B, C

* In part b of both #19 and #22 you were asked to run Nearest Neighbor from only **two** additional vertices. All of them are worked below.

Problem 19

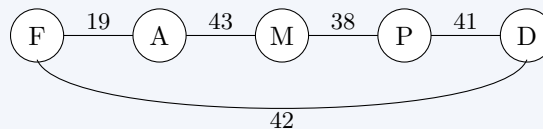
A company needs to deliver product to each of their 5 stores around the Dallas, TX area. Driving distances between the stores are shown below. Find a route for the driver to follow, returning to the distribution center in Fort Worth:

- Using Nearest Neighbor starting in Fort Worth
- Using Repeated Nearest Neighbor
- Using Sorted Edges

	Plano	Mesquite	Arlington	Denton
Fort Worth	54	52	19	42
Plano		38	53	41
Mesquite			43	56
Arlington				50

(a)

Solution

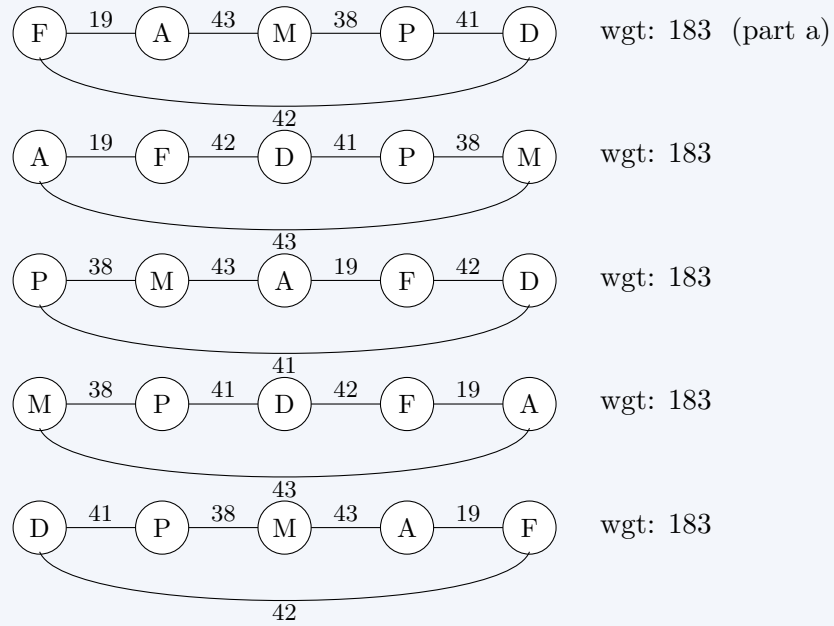


Weight:

(b)

Solution

Starting vertex:



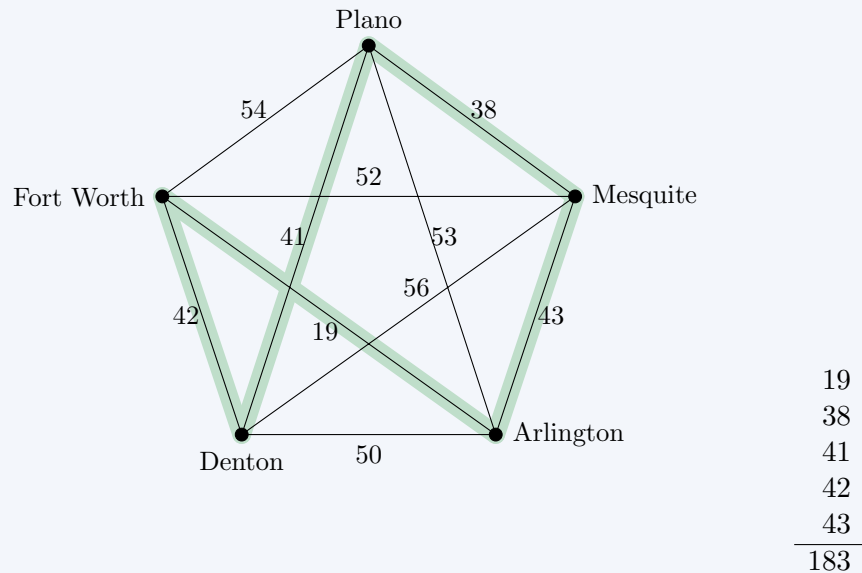
ANS: Pick $F-A-M-P-D-F$, wgt 183.

Every start produces the same circuit, just entered at a different point, so all five weigh 183.

(c)

Solution

Sorted Edges:



The edges are taken in order of weight: 19 ($FW-A$), 38 ($P-M$), 41 ($P-D$), 42 ($FW-D$), and finally 43 ($M-A$), which closes the circuit.

This is the same circuit as in parts (a) and (b), so all three algorithms agree here.

Problem 22

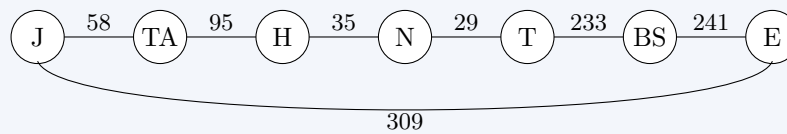
A tourist wants to visit 7 cities in Israel. Driving distances, in kilometers, between the cities are shown below. Find a route for the person to follow, returning to the starting city:

- Using Nearest Neighbor starting in Jerusalem
- Using Repeated Nearest Neighbor
- Using Sorted Edges

	Jerusalem	Tel Aviv	Haifa	Tiberias	Beer Sheba	Eilat
Jerusalem	–	58	151	152	81	309
Tel Aviv	58	–	95	134	105	346
Haifa	151	95	–	69	197	438
Tiberias	152	134	69	–	233	405
Beer Sheba	81	105	197	233	–	241
Eilat	309	346	438	405	241	–
Nazareth	131	102	35	29	207	488

(a)

Solution



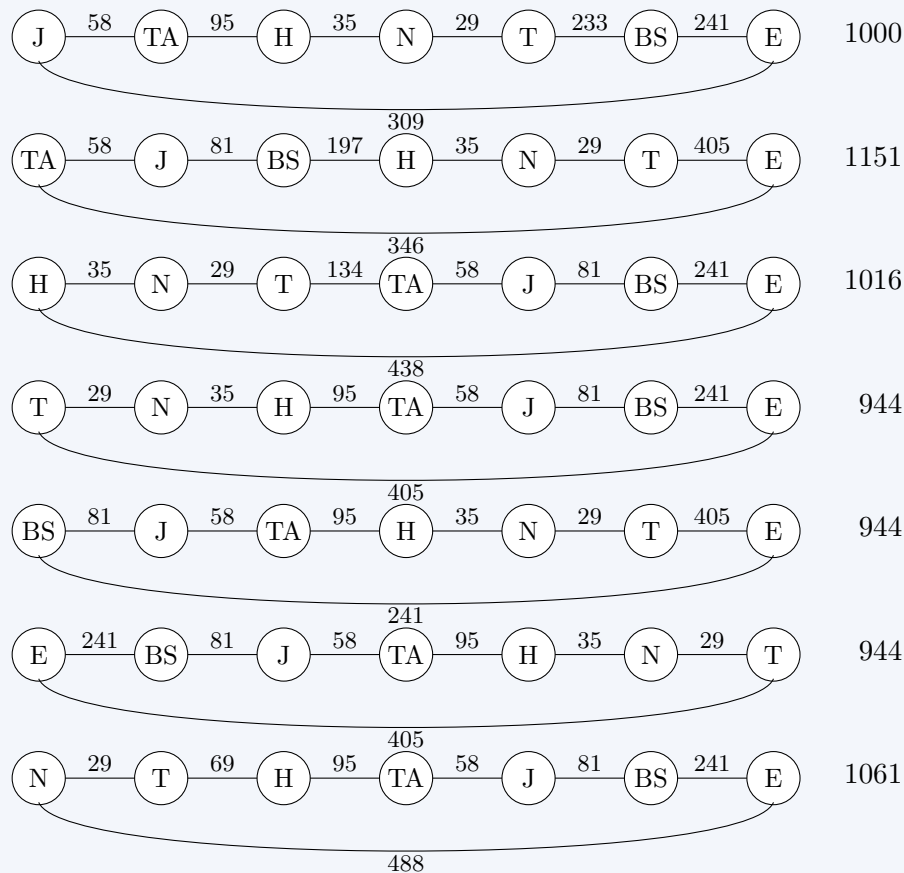
The six edges along the top come to 691.

Adding the return leg $E-J = 309$ gives a circuit of weight 1000.

(b)

Solution

Starting vertex, and weight:

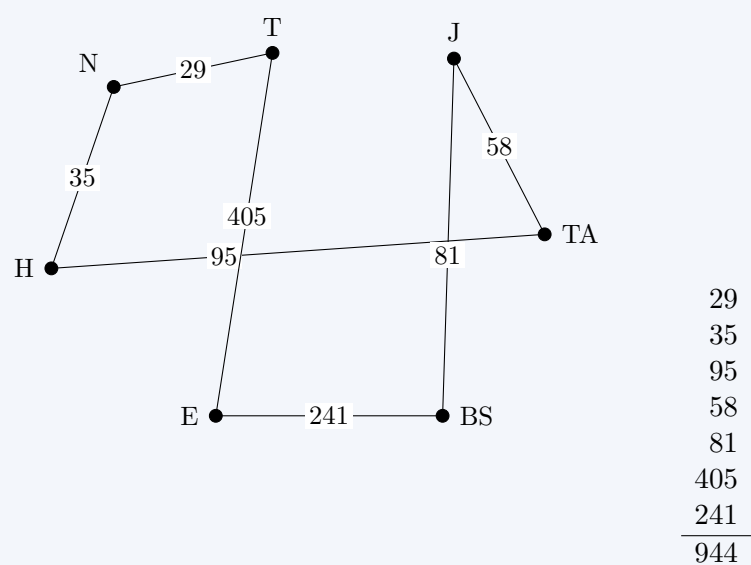


Pick $T-N-H-TA-J-BS-E$, wght: 944

Tiberias, Beer Sheba, and Eilat all give a circuit with weight 944 because they produce the same circuit entered at three different starting points.

(c)

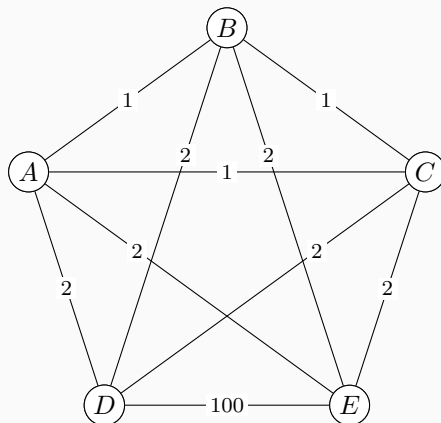
Solution



This is the same circuit that part (b) found from Tiberias, Beer Sheba and Eilat.

Problem A

Answer questions about the graph below.

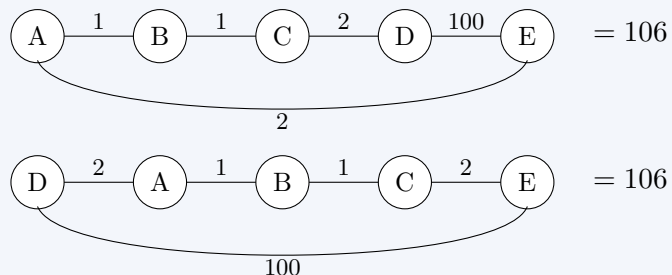


1. Apply the Repeated Nearest Neighbor Algorithm and determine the weight of the resulting circuit. (Note that you are allowed to use symmetry in the graph to make this faster!)
2. Apply the Sorted Edges Algorithm and determine the weight of the resulting circuit.
3. Using your own judgment, find a minimum weight Hamiltonian circuit.
4. Does either algorithm — Repeated Nearest Neighbor or Sorted Edges — return a minimum weight Hamiltonian circuit?

(1)

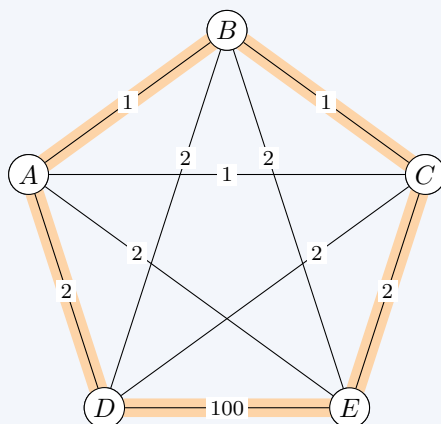
Solution

Using symmetry, we see that RNN gives the same weight starting at A , B , or C , and D and E will be the same:



(2)

Solution

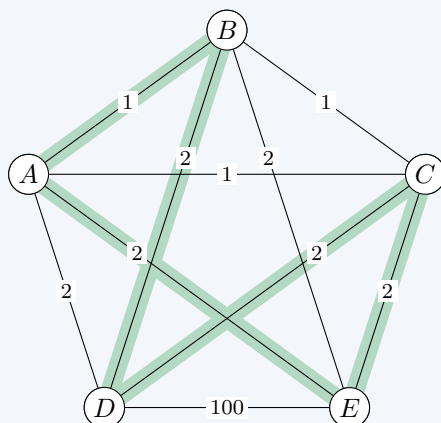


Weight 106

Sorted Edges takes AB and BC at weight 1 and then must skip AC , which would close a triangle. Of the weight-2 edges it takes AD and CE , and the circuit can only be closed with $DE = 100$.

(3)

Solution



Weight = 9

Any circuit has to give D two edges and E two edges. Avoiding the weight-100 edge between them means each takes two weight-2 edges, for 8, and the fifth edge can then be one of the weight-1 edges — so 9 cannot be beaten.

(4)

Solution

No. RNN and SE do not give an optimal solution.

Both algorithms take the cheap edges among A , B and C first and are then forced to pay the 100 to get from D to E ; the optimal circuit gives that up front by keeping D and E apart.

Problem B

Suppose someone wants to visit the capital city of every state in the contiguous 48 states and Washington DC. So, they will visit 49 cities in total. [NOTE: you will need to use a computational tool, like WolframAlpha, to complete this problem.]

1. Suppose they want to start and end the 49-city-tour at the same place. How many different tours are possible? (Your answer should be in *both* factorial notation and in scientific notation.)
2. Suppose you have a computer that can calculate the length of 1000 49-city-tours in one second. How long would it take the computer to calculate the length of all possible 49-city-tours? Give your answer in *years*.
3. What does your answer in part 2 indicate about the Brute Force algorithm?

(1)

Solution

$$(48)! = 1.24 \times 10^{61}$$

Fixing the starting city leaves the other 48 to be ordered, and each tour is counted once this way.

(2)

Solution

$$\frac{1000}{\text{sec}} \cdot \frac{60 \text{ sec}}{\text{min}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{24 \text{ hr}}{1 \text{ day}} \cdot \frac{365 \text{ days}}{\text{year}} = 3.1536 \times 10^{10} \text{ per year}$$

So

$$\frac{(48!)}{1000 \cdot 60 \cdot 60 \cdot 24 \cdot 365} = 3.93 \times 10^{50} \text{ years} !!$$

(3)

Solution

Even with a computer, Brute Force will take too long.

Problem C

A company plans to run a bus continually throughout the day in a loop around 5 tourist stops in a city, denoted A, B, C, D, and E. The company would like to find a route that minimizes the time to complete a loop. Driving times in minutes are shown in the table.

Driving Times in Minutes					
	A	B	C	D	E
A	–	4	7	3	2
B	4	–	5	6	9
C	7	5	–	8	6
D	3	6	8	–	5
E	2	9	6	5	–

Using terminology from Graph Theory, what is the company looking for?

Solution

The table gives a driving time for every pair of stops, so it describes a complete weighted graph on 5 vertices: each tourist stop is a vertex, each edge is the trip between two stops, and the weight of an edge is the driving time in minutes.

The bus has to call at every stop and come back to where it started, which is a Hamiltonian circuit, and the company wants the one that takes the least total time. So they are looking for a

minimum weight Hamiltonian circuit in this graph.

(This is the Traveling Salesman Problem.)